

Artificial Intelligence for ETF Market Prediction and Portfolio Optimization

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Abstract

In asset allocation and time-series forecasting studies, few have shed light on using the different machine learning and deep learning models to verify the difference in the result of investment returns and optimal asset allocation. To fill this research gap, we develop a robo-advisor with different machine learning and deep learning forecasting methodologies and utilize the forecasting result of the portfolio optimization model to support our investors in making decisions. This research integrated several dimensions of technologies, which contain machine learning, data analytics, and portfolio optimization. We focused on developing robo-advisor framework and utilized algorithms by integrating machine learning and deep learning approaches with the portfolio optimization algorithm by using our predicted trends and results to replace the historical data and investor views. We eliminate the extreme fluctuation to maintain our trading within the acceptable risk coefficient. Accordingly, we can minimize the investment risk and reach a relatively stable return. We compared different algorithms and found that the F1 score of the model prediction significantly affects the result of the optimized portfolio. We used our deep learning model with the highest winning rate and leveraged the prediction result with the portfolio optimization algorithm to reach 12% of annual return, which outperform our benchmark index 0050.TW and the optimized portfolio with the integration of historical data.

Keywords: artificial intelligent (AI), deep learning, exchange traded funds (ETF), financial market prediction, machine learning, portfolio optimization

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1. Introduction

The flourishing of financial technology (Fintech) and deep learning has received enormous attention from a wide spectrum of industries in recent years. The World Economic Forum (2015) identified fintech as a disruptive innovation; the disruptive power of fintech innovations will manifest clearly as the market evolves [1]. In 2012, Hinton and Krizhevsky proposed AlexNet to prove the strong capability of deep learning in the ImageNet competition. The result shows that convolution neural network (CNN) had outperformed significantly than its traditional counterpart. Subsequently, artificial intelligence and deep learning have rapidly changed nearly every industry.

For stock market forecasting, deep learning model, recurrent neural network with time series forecasting, and attention mechanism have provided artificial neural networks the capability to compute the relationship between time and trends, and the outcome of these successes is to contribute additional possibilities to predict the future and make good decisions. Section 2 will discuss in detail the relevant literature.

In recent years, fintech has received enormous attention in science, information technology, and financial industries. Using different types of strategies for stock market prediction has become a trend in the trading market. Some studies investigate on portfolio optimization models based on several traditional methods; however, little research combines deep learning and asset allocation. Moreover, a wide range of improvement remains in the precision of forecasting. Therefore, this study focuses on using machine learning and deep learning model for stock market forecasting; then it integrates theories of portfolio optimization to contribute the additional return of investment (ROI) to customers by accepting our portfolio recommendations.

This study aims to use artificial intelligence technology, including different machine learning and deep learning mechanisms, to forecast the trend of ETFs listed in Taiwan and US markets. Moreover, we will compare the performance of different machine learning and deep learning models to forecast the investment returns and utilize the best practice of forecasting data to allocate our future portfolio. We use two different asset allocation theories to optimize our portfolio. Furthermore, by comparing buy and hold or average weighted strategies, we can determine which theories perform better against others.

Table 1. A review of related works on stock market prediction

Authors(Years)	Data Type	Methods	Accuracy
Jui-Sheng Chou et al. (2018)	S&P 500 index from Oct 5, 2011 to May 31,2017	SVM	45.48%
Magnus Hansson et al. (2017)	S&P 500 index from 2009-01-02 to 2017-04-28	ARMA	50.43%
Magnus Hansson et al. (2017)	S&P 500 index from 2009-01-02 to 2017-04-28	Deep-LSTM	50.72%
Magnus Hansson et al. (2017)	S&P 500 index from 2009-01-02 to 2017-04-28	LSTM	51.88%
Yang Jiao et al. (2017)	S&P 500 index from Jan 2, 2009 End Jun 30, 2017	Random Forest	58.00%
Chang Sim et al. (2013)	S&P 500 index from 1 January 2000 to 31 December 2009	ANN	56.30%
Huacheng Wang et al. (2009)	Shanghai Stock Exchange form 2000 to 2003	Decision Tree	54.00%
Huacheng Wang et al. (2009)	Shanghai Stock Exchange form 2000 to 2003	Bagging-decision tree	61.87%
Huacheng Wang et al. (2009)	Shanghai Stock Exchange form 2000 to 2003	Boosting-decision tree	59.16%

This research then aims to propose the architecture for robo-advisor to deliver different investment consultation services for short- and long-term periods. The proposed robo-advisor integrated several dimensions of technologies, which contain machine learning, data analytics, and portfolio optimization. The result of our research can be applied into various applications, such as comparing different algorithms and determining the better practice to support decision-making, and providing investment recommendation to increase the ROI of the optimized portfolio.

The research questions are as follows:

RQ1: Whether the combination of deep learning and investing strategy can improve effectively the return of investment.

RQ2: Compare the returns among different assets allocation theories to find which strategies outperform others.

The research value for this study is that we hope that the result of this study will become a submodule of robo-advisor and financial recommendation system. We recommend the investment strategy to investors, assist them in reaching a high return of investment, and according to the investors' opinion and risk tolerance coefficient, we can customize the portfolio recommendation to each type of investors. The prediction generated by our deep learning and machine learning models can serve as a decision-making support system. According to different terms of investment, we provide different techniques to forecast the trends to meet each investor's requirement and expectation. We hope this achievement will bring many intelligent services to the financial community and use cutting-edge technology to enable a highly progressive and flourishing financial industry.

The rest of this paper is structured as follows. Section 2 describes the literature review on assets allocation, deep learning and machine learning. Section 3 establishes the research methodology and design. Section 4 shows the experimental results and discussion. Finally, Section 5 presents the conclusions of this research.

2. Related Work

2.1. Artificial Intelligence in Finance

Statistical analysis based on historical trends prediction, such as the autoregressive integrated moving average (ARIMA) model [2], the autoregressive conditional heteroscedasticity (ARCH) model [3] model, and the generalized autoregressive conditional heteroscedasticity (GARCH) model [4], has been extensively utilized to predict the financial market. However, these models do not perform well owing to their limitations. Accordingly, the current research will propose machine learning and deep learning models and attempt to solve the issues.

In recent years, implementing the machine learning model to forecast the stock trend has become the trend in stock market, and some discussions appear in Chen et al. [5]; they proposed that deep learning model indeed provides a certain benefit of stock prediction but remains a complicated task because of the factors, such as the economy, politics, environment, and culture; many chaotic indicators remain, which will affect the financial market and the forecasting capability.

2.2. Machine Learning and Deep Learning

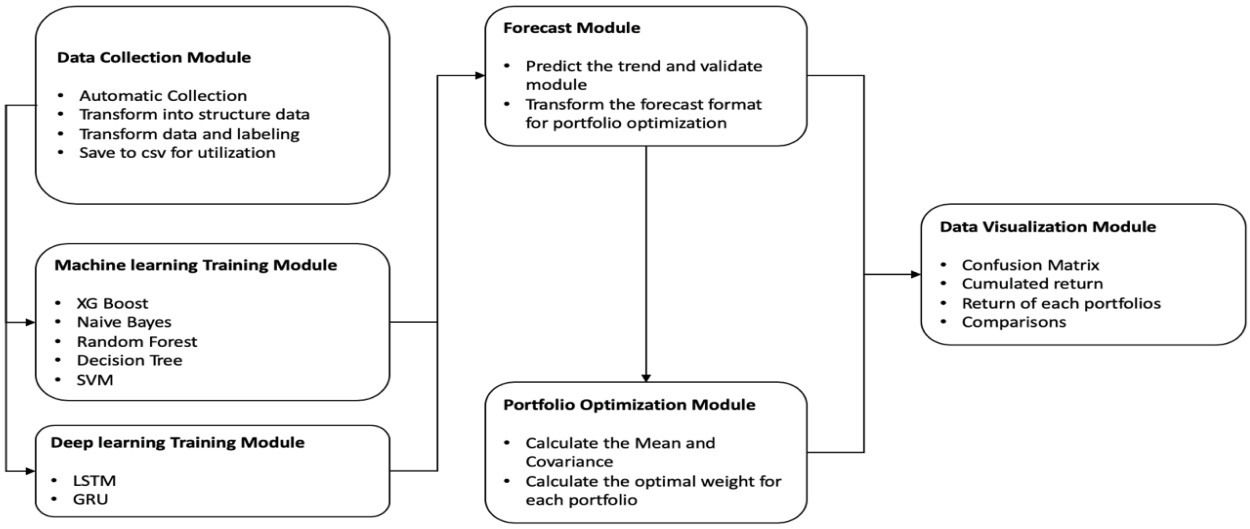


Figure 1. System architecture of the proposed robo-advisor

Machine learning is a field of computer science that uses different statistical techniques to grant computer systems the capability to "learn" things with data, without undergoing explicit programming [6]. The current research will leverage five machine learning models to generate investor's view for the portfolio optimization model, The model includes support vector machine (SVM) [7], Decision Tree [8], eXtreme Gradient Boosting (XGBoost) [9], Random Forest [10], and Naïve Bayes [11].

Deep learning is a branch of machine learning, specifically known as multiple hidden layer neural networks. In addition, it has some advantages of performing multi-dimensional calculation and the capability to calculate the weight of each parameter automatically to determine the minimum loss and highest accuracy. The concept of deep learning has been introduced since the 1990s. However, the efficacy of hardware and the efficiency of computing were insufficient for running deep learning models and reaching high performances; deep learning started flourishing with the proposal of AlexNet [12] and Deep Learning [13]. Table 1 shows that the empirical studies about different methodologies in applying machine leaning mechanism to predict stock movement; those studies use accuracy index to validate the model and evaluate the ability of prediction. Machine learning and deep learning models reach the accuracy of 50% to 60 %, respectively. The current research implements the LSTM [14] and GRU [15] as its forecast model. This research will compare the performance of different models and optimize the testing accuracy by tuning and modifying the parameters.

2.3. Portfolio Optimization

In 1952, Harry Markowitz introduced the earliest study of portfolio and assets allocation known as the Markowitz Mean-Variance Model [16]. However, the deviation will be enlarged when subjected to the highly sensitive input parameters. Thus, the Black-Litterman model proposed by Black and Litterman in 1992 has been derived [17] to address the disadvantage.

3. Research Methods and System Framework

This section will depict our research methods to propose our system framework of day trading module and portfolio optimization module. The first section describes the research design and the research structure of this paper; the second section describes the system architecture; the third section describes the data collection and the subject of the data; the fourth section proposes the supervised machine learning transformation and data normalization. The fifth section describes the implementation of time series deep learning methods and different machine learning methodologies.

3.1. Research Methodology and Design

Our research method adopts the system development research methodology in the field of information system research (System Development Research Methodology) [18]. This methodology has been divided into the five stages of our day trading and portfolio optimization system.

3.2. System Architecture

Figure 1 presents the system architecture of the proposed robo-advisor. To propose the time series forecasting task, we develop several modules for our system. To perform machine learning and deep learning forecasting, we use Scikit-learn as our library, which a flexible and powerful library for implementing machine learning models. In deep learning time series forecasting, we use Keras which is known as the high-level deep learning API executed on top of the TensorFlow, for developing our LSTM and GRU model.

3.3. Data Collection and Data

We use the exchange traded funds (ETFs) listed in the Taiwan market as our subjects, and we use the daily adjusted close price as input variable, and the target variable is the trend of next day. The period of our stock collection is approximately 11 years from 2008/01/02 to 2018/12/31, with a total of 2,768 trading days. From 2018, we validate approximately 245 trading days. We collected 20 ETFs listed in the Taiwan market. Our data were automatically obtained from data collection module, sourced by the Yahoo Finance API.

3.4. Data Transformation and Normalization

In supervised machine and deep learning, we have to transform the data into a time series supervised learning format. We set up a moving window of the past 120 trading days for the day we are attempting to make a prediction, meaning that 120 features exist for each row. Moreover, we use the highly complicated data to capture the movement of time series, We use L2 normalization to the input data for ease of computing. For example, on 2017/12/28, the data from 2017/07/11 to 2017/12/27 (totaling 120 trading days) will be set as our input data, and we will normalize the data to reach a good performance.

3.5. Models Training and Validation

We split the entire dataset into a training dataset and a testing dataset. The time interval of our training dataset starts from January 2008 to December 2017. The testing dataset continues from January 2018 to December 2018. We use the confusing matrix to illustrate the difference between predicted results and the actual trends. We leverage the Python library Scikit-learn to implement machine learning algorithms, while the high level library Keras with TensorFlow is used to build up deep learning model.

4. Experimental Results and Discussions

4.1. Model-based Day Trading Strategy

We leverage a day trading investing strategy in our module by using the forecast result we obtained from different machine and deep learning models. We assigned our output data a 1 (goes up) and -1 (goes down) to signify the trend of next trading day. We conduct this trading strategy within the trend prediction section on each trading day in 2018 and used the F1 score as an indicator to validate the model performance of our strategy. We set up a buy and hold strategy as our control group, which means that we purchase the target stock at the beginning of 2018 and hold it until the end of the year.

4.2. Results and Analysis

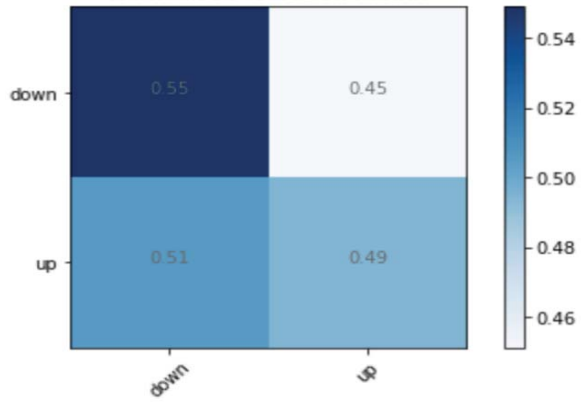


Figure 2: Confusion matrixes for 0050.TW prediction in 2018

Table 2: Predicted result of 0050.TW by machine learning model

Trend	precision	recall	F1-score	support
Down	0.55	0.35	0.42	130
Up	0.49	0.69	0.58	120
Average	0.52	0.51	0.50	250

In this section, we present the results of forecasted cumulative of each model. Figure 2 presents the confusion matrixes for 0050.TW prediction in 2018. It shows the capability of each prediction model for each EFT, to demonstrate whether our model can derive better performance in each asset.

Table 2 shows the performance of our machine learning model by forecasting the trend of 2018 in the entire years (250 trading days); we do not only evaluate the model by accuracy, but we also observe our model by the precision, recall, and the F1-score. Our model shows that the prediction result is highly objective and not distracted by the unequal portion of dataset. We train the model by modifying the parameter and find the best practice of each implementation through manual modification and grid search optimization to reach the best performance. The grid search can help us identify the best parameter setting by training multi-modals and propose the best model of a set of parameters called the best estimator. We use the best estimator of each model to find and elevate the F1 score.

4.3. Comparison with Longer Time Horizon

Our models are trained with the data from 2008 to 2017, and tested the data in 2018. We use 0050.TW as our example equity and use the SVM as our training model. Figures 3 and 4 show that the volatility and the cumulated return of the forecasting result generated by SVM model. We set up three different comparison methods, including buy and hold, results generated by SVM model, and the baseline of doing nothing. In the cumulated returns, the figure can tell the differentiation of returns between forecasted result and buy and hold strategy. Such figure

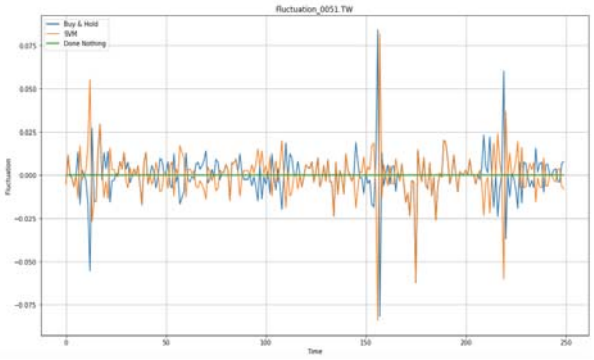


Figure 3: Volatility for 0050 TW prediction in 2018



Figure 4: Performance of investment strategies

Table 3: Predicted results of 0050.TW by machine and deep Learning models

Model	Winning Rate	% Better than Buy and Hold Strategy
LSTM	**60%**	**25%**
GRU	50%	15%
RF	45%	10%
DT	40%	5%
NB	40%	5%
SVM	50%	15%
XGB	55%	20%

Note. RF = Random Forest; DT = Decision Tree; NB = Naive-Bayes; XGB : XGBoost

also shows that the predicted result is better than the buy and hold strategy.

Table 3 shows the predicted results of 0050.TW by machine and deep learning model. We selected the best winning rate model to predict the result of each asset. Then, we transform the prediction result to the portfolio optimization section and noted that the winning rate of buy and hold strategy is 35%. To illustrate, 7 of the 20 stocks obtain a positive cumulative return in 2018; thus, the winning rate of the baseline is 35%. The forecast results generated by LSTM model can reach 12 of 20 with positive

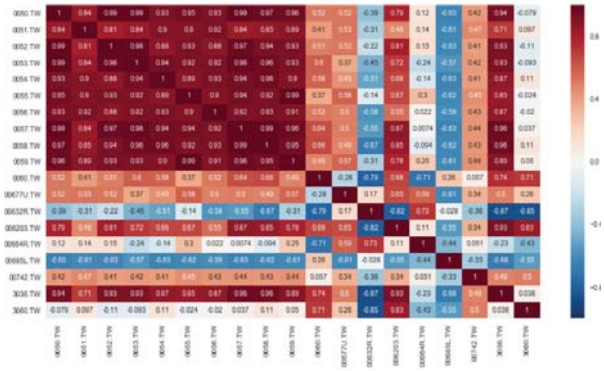


Figure 5: Correlation matrix for 20 ETFs

cumulative return; thus, the winning rate of LSTM is 60% and outperforms the baseline by 25%.

4.4. Portfolio Assets Selection

Figure 5 shows that the correlation among portfolio assets. We calculate the adjusted close price by group and illustrate the correlation matrix. We found six equities, which have relatively low correlation to construct our portfolio. Table 4 shows the correlation matrix for the selected six ETFs. In this table, we can tell the correlation coefficients among selected equities and the negative correlations between them. Thus, we can reduce the risk by proposing this portfolio. We expect that result of our portfolio assets selection can reduce the risk and obtain the best performance when we implement the portfolio optimization models.

4.5. Portfolio Optimization Using the Model-based Investor Views

We leverage the portfolio optimization model to demonstrate asset allocation via the combination of investor views, which is derived from our forecasting model and transform the predicted result to our portfolio optimization model as input parameters. We have chosen six assets with low correlation coefficient to construct our portfolio. Table 5 presents the list of selected six ETFs. We have quarterly adjusted the formation of our portfolio to reach the best performance and high cumulative return. We set up the 0050.TW as a benchmark of our model, which can show the comparison result with high objectivity because of its stable performance. We then explain to our investors the advantages and the good performance of our system.

4.6. Portfolio Optimization and Performance Comparison

Following the steps, we implement the Black-Litterman model to optimize our portfolio. For the optimization process, we must prepare the data, which include historical average returns and the covariance matrix (the correlation among each chosen stock) for the Black-Litterman

Table 4: Correlation matrix for the selected six ETF

	0050	00677U.TW	00632R.TW	006203.TW	00664R.TW	00685L.TW
0050	1.000000	0.004479	0.047665	-0.067147	-0.012356	-0.036473
00677U.TW	0.004479	1.000000	-0.079317	-0.037255	-0.020727	-0.058058
00632R.TW	0.047665	-0.079317	1.000000	-0.026025	-0.028053	0.239930
006203.TW	-0.067147	-0.037255	-0.026025	1.000000	-0.002381	-0.108372
00664R.TW	-0.012356	-0.020727	-0.028053	-0.002381	1.000000	-0.015700
00685L.TW	-0.036473	-0.058058	0.239930	-0.108372	-0.015700	1.000000

Table 5: List of selected six ETFs

Symbol	Name
0050.TW	Yuanta/P-shares Taiwan Top 50 ETF
00677U.TW	Fubon S&P 500 VIX Short-Term Futures ETF
00632R.TW	Yuanta Daily Taiwan 50 Bear -1X ETF
006203.TW	Yuanta/P-shares MSCI Taiwan ETF
00664R.TW	Cathay TAIEX Daily Inversed ETF
00685L.TW	Capital TAIEX Daily Leveraged 2X ETF

Table 6: Weight of six ETFs of the Black-Litterman model

Symbol	Weight
0050.TW	0.262296
00677U.TW	0.039492
00632R.TW	0.051052
006203.TW	0.125719
00664R.TW	0.232864
00685L.TW	0.288578

portfolio optimization model. The next step involves our implementation of the investor views, which means that we use our prediction model to predict each stock and define the disparity among each stock performance. We then combine with the historical returns and generate the posterior mean and posterior covariance matrix by applying the Black-Litterman formula. We leverage the Markowitz mean-variance optimization model for another portfolio optimization strategy. The model is sensitive to the investor views, and the entire model might be affected by the weight manipulation of a single stock. The latter might cause substantial fluctuations and unpredictable result. The weight of the Markowitz portfolio is generated from the mean-variance optimization model by using quarterly historical returns and the covariance matrix. Table 6 presents the weight of six ETFs of the Black-Litterman Model.

Figure 6 presents the cumulative returns of our selected stocks, and the time interval is the entire year of 2018, which is approximately to 245 trading days. The cumulative return of each selected stock has a normal performance, and the cumulative return of our portfolio is approximately equal to zero. In 2018, most of the stocks have no remarkable performance in the stock market; most of the stocks have a normal or even negative performance. The current research shall forecast the stock market, reach the performance, and use the portfolio optimization methodology to decrease the volatility while obtaining a highly stable performance.

Figure 7 presents the result of the portfolio optimization derived from our forecasting model. We use the long short-term memory to predict the trend of each stock and utilize the correlation matrix to propose our portfolio. We select six stocks of negative correlation coefficient to construct our portfolio, and we implement the return of each trading days and transform into the Black-Litterman Model. We demonstrate that the model can propose a better performance than the actual return.

The dramatic differentiation between Figure 6 and 7 is apparent. Figure 7 shows the actual return and implements the same portfolio optimization algorithm; the return of portfolio can reach approximately reach zero, although our customers will not obtain a deficit in the portfolio with actual data. However, they still cannot gain any profit from the portfolio for the entire year (but still attain a better performance than the 0050.TW). Conversely, Figure 7 shows that the result derived from our prediction model can reach more than 12% of profit by utilizing the Markowitz mean-variance model and average weight strategies. Moreover, the relatively stable Black-Litterman Model can still gain more than 7% return in the bearish market. The result shows that the portfolio with forecast data input outperforms the 0050.TW and even the optimized portfolio with actual data.

5. Conclusions

This section will focus on the research finding of this study, the contribution of this research and practitioners' implication, and how it benefits the real world when addressing the existing issues or problems. Furthermore, we will shed light on the future research for improving our performance based on this study. Last but not least, we depict our future vision by elaborating the integration with other system or module to elevate its usefulness in meeting the market's needs.

In this study, we have developed a robo-advisor with different machine learning and deep learning forecasting methodologies and utilize the forecasting result of the portfolio optimization model to support our investors in making decisions. We have integrated several dimensions



Figure 6: Cumulative returns of portfolio optimization strategies (Returns from actual return)

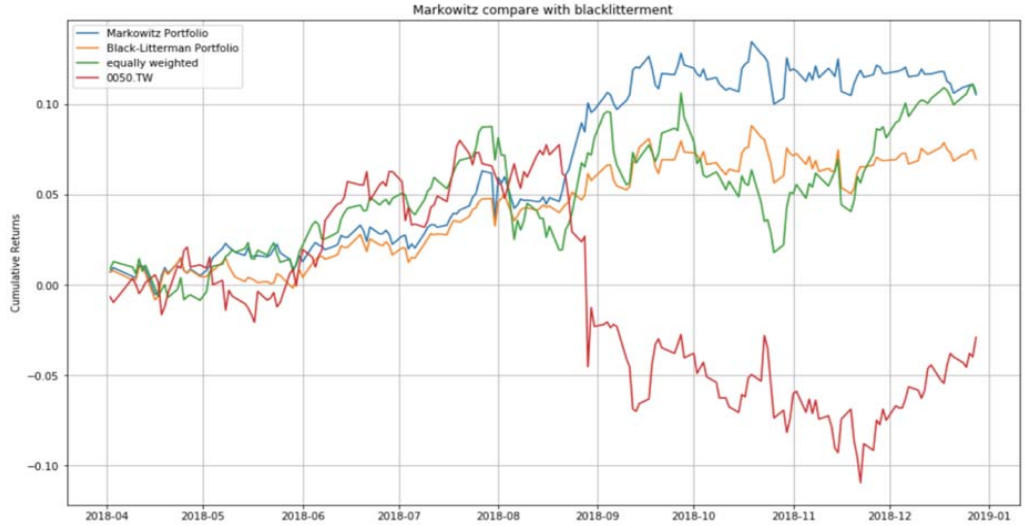


Figure 7: Cumulative returns of portfolio optimization strategies (Returns from prediction model)

of technologies, which contain machine learning, data analytics, and portfolio optimization. We focused on developing robo-advisor framework and utilized algorithms by integrating machine learning and deep learning approaches with the portfolio optimization algorithm by using our predicted trends and results to replace the historical data and investor views.

5.1. Research contributions

The contributions of this research are as follows:

- The proposed system can be integrated into the portfolio optimization robo-advisor recommendation system as a sub-module to assist the investors in making long- or short-term investment decision.
- Our results suggest that LSTM deep learning model performs better than machine learning models. The recurrent neural networks have a better computation capability, specifically for time series data. Moreover, the stateful LSTM can benefit more reliable stock forecasting result than the traditional or ensemble machine learning methods when processing and forecasting the day trading data.
- In this research, we proposed the portfolio optimization model and integrate deep learning forecasting. In the neutral stock market, we can still outperform the baseline ETF and attain a better performance and the optimized portfolio with the actual data implementation.

5.2. Recommendations for Future Research

In the future, we can improve our model with following methods :

- Integrate additional technical indicators for our model training. When fed with various inputs, the model can extract important features during the model training, thus determining important indicators.
- Integrate our forecasting model with natural language processing technology and combine the investors' comments and opinions with the stock market daily price to determine whether the comments from the Internet and social media genuinely represent future trends of stock market.

6. Acknowledgment

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